In Silico Dynamics of Quorum Sensing

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1. Purpose of this document and some introduction

This should give an slight overview of Quorum Sensing (QS) and the model/code that I wrote.

Here is a link to the github. Most of the calculations use linear algebra and vectorized operations to speed up computations/use numpy's speed. Spring 25 exploration contains the most up to date code. There is some past stuff, including a python version of Wang's original code available as well. In the Json folders are the results of runs, All the json files in spring 25 exploration contain a key called "params" which has the starting parameters of simulation for some reproducibility.

QS is an important evolutionary mechanism to study as it is a form of communication at some of the most basic life forms, thus studying the dynamics of its evolution could contribute to understanding how communication evolves in general. As for why *in silico* is important to study, if we can replicate experimental data in a controlled mathematical environment, this will better allow us to understand the governing dynamics of how communication evolved. What I mean by this is that if we know we can replicate experimental data using some fitness evaluation, we can then apply that knowledge to other known environments to see if communication evolved in a similar pathway to QS, it would also allow us to assess the viability of communication in certain environments, and if one seems unfavorable and communication did evolve, it suggests that there are multiple pathways that can govern it's evolution.

We have the following assumptions that where used to justify the creation of the model:

- A signal type exists
- In propagule pool, cellular heterogeneity can be obtained via mutation and selection
- Each sub-population is founded a set number of individuals
- We can prescribe a fitness to each individual using calculated values
- Mass transfer is a constant factor in each testing environment.
- The signal causes a reaction to produce an enzyme that then causes the cell to then benefit from a third environmental molecule.
- Signal concentration in each environment rapidly approached equilibrium estimated using differential equations.
- Relative cost of cooperation is quantifiable, and the benifit associated with cooperation is quantifiable and a function of the cost and density.
- The overall benifit should be higher at higher densities and minimally impactful at lower ones.

2. The basis of the model

- Signal(S) Expression

$$\frac{dS}{dt} = p\left(1 + r\frac{S}{K_S + S}\right)N - (u+m)S$$

$$S^* = \frac{\sqrt{(K_S(u+m) - N(pr+p))^2 + 4K_SNp(u+m)} - K_S(u+m) + N(pr+p)}{2(u+m)}$$

  - p - production value
  - r - autoinduction value
  - K_S - half signal concentration
  - N - cellular density
  - u - decay rate
  - m - mass transfer

- Enzyme(X) Secreted

$$\frac{dX}{dt} = \eta\left(\frac{S}{S + K_X}\right)N - (m+f)X$$

$$X^*(S^*) = \frac{N\eta\left(\frac{S^*}{S^*+K_X}\right)}{m+f}$$

  - η - max production value
  - K_X - half signal concentration
  - N - cellular density
  - f - decay rate
  - m - mass transfer

- Beneficial molecule(Y) interaction

$$\frac{dY}{dt} = \xi\frac{X}{X + K_Y} - (cN + m + e)Y$$

$$Y^*(X^*) = \frac{\xi\frac{X^*}{X^*+K_Y}}{(cN+m+e)}$$

  - ξ - rate of interaction between X and Y
  - c - rate of consumption
  - K_Y - half signal concentration
  - e - decay rate
  - m - mass transfer

3. Fitness Calculation of ith individual

$$F_i = B_0 + B_{coop} \sum_{N_j, m_k} Y^*(X^*(S^*)) - C_{coop} \sum_{N_j, m_k} X^*(S^*) - C_{sig} \sum_{N_j, m_k} S^*$$

- B_0 - Baseline fitness
- B_{coop} - Benefit of communication
- C_{coop} - Cost of communication
- C_{sig} - Cost of signaling
- N_j - cellular densities being tested
- m_k - mass transfer being tested
- $$S^* = \frac{\sqrt{(K_S(u+m) - N(pr-p))^2 + 4K_S Np(u+m)} - K_S(u+m) + N(pr+p)}{2(u+m)}$$
- $$X^*(S^*) = \frac{N\eta\left(\frac{S^*}{S^*+K_X}\right)}{m+f}$$
- $$Y^*(X^*) = \frac{\xi \frac{X^*}{X^*+K_Y}}{(cN+m+e)}$$

4. Model with evolutionary wrapper

- Evolving parameters for the signal (max, min)
  - p_i - production rate $(0, 5e-07)$
  - η_i - X production rate $(0, 5e-07)$
- Environmental testing parameters
  - N_j - cellular densities (1e1.5 to 1e5 100 steps)
  - m_k - decay rates (10.0 ** -7.0 to 1e-4 100 steps)
- Constant parameters
  - B_0 - Baseline fitness (100)
  - B_{coop} - Benefit of communication (0.0015)
  - C_{coop} - Cost of communication (0.005)
  - C_{sig} - Cost of signaling (0.00032)

– r_i - auto induction ration - 5
– u_i - decay rate $4e-06$
– K_S, K_X, K_Y - half signal concentration (50, 500, 500)
– f - decay rate for X (4e-06)
– ξ - rate of interaction between X and Y (2.0)
– c - rate of consumption of Y (1e-8)
– e - decay rate for Y (4e-06)

1. Initialize population of $size\ pop$ with each individual i with p_i, η_i
2. Calculate fitness F_i for each individual and store as $F = (F_1, ..., F_{size\ pop})$.
3. Select a new population using the normalized fitness vector, $\frac{F}{||F||_{L^1}}$, as the probability distribution with replacement.
4. For each evolving parameter
   (a) Randomly select a subset to mutate
   (b) Mutate with truncated normal distribution with constant standard deviation, min, and max. Use the individual's value as the mean.
5. Repeat 2-4 until the desired number of generations is reached

5. Cellular Density and Mass transfer impact

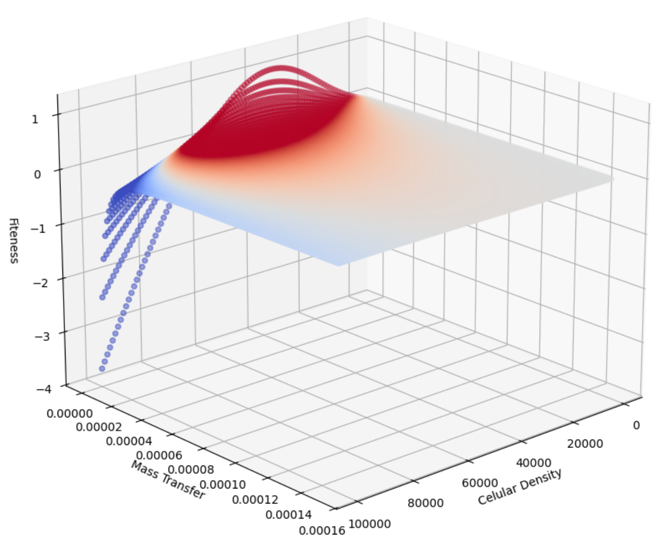

Figure 1: This is an example of an optimal fitness function. The color is just the fitness (z-axis) to better see the shape), red is positive impact and value is negative impact. Impact of Cellular Density and Mass Transfer on Fitness. This is taken at the maximal p, η values in section 5.

6. Optimization

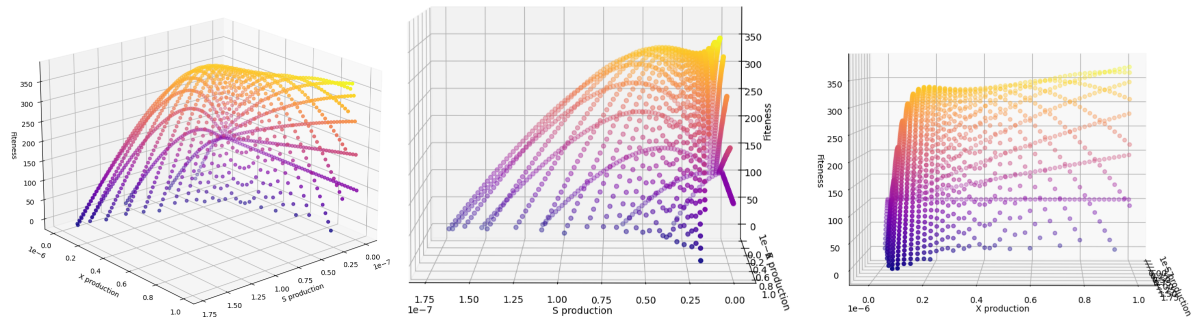

Figure 2: The color is just the fitness (z-axis) to better see the shape). It seems that after about $\eta = 2e - 7$, we get there is a small stable local max for each p. I think this is due to the fact that, S is basically saturated at this point and X is becoming saturated as well.

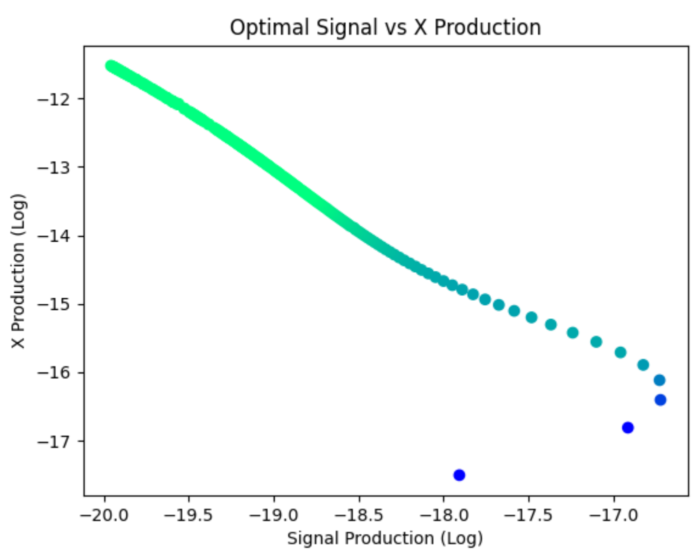

Figure 3: For each η, I found the numerical best p. Here the color represents max fitness, the lighter the more fit. Each scale is log. Here the log scales have a pretty good correlation however the exact relationship I am unsure about.

7. Results of Some testing

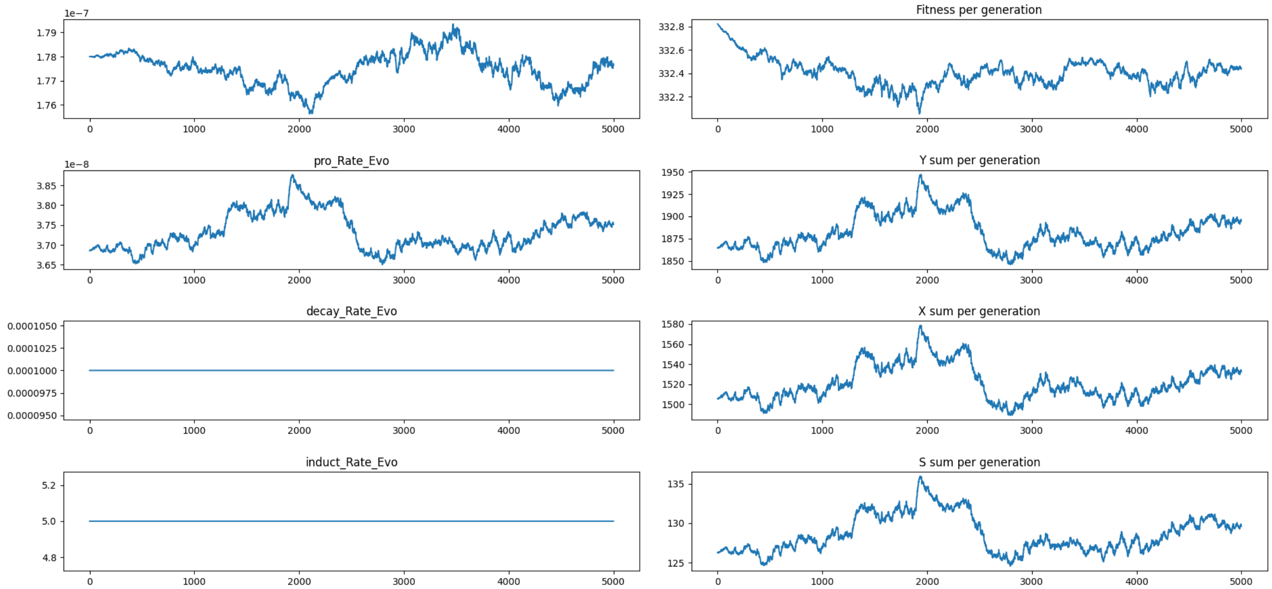

Figure 4: This is the result of a run starting at the optimal fitness. We can see that the fitness actually decreases. It seems like it wants to decrease the X production rate while increasing the production rate, this might be going to a place that has a better topography, i.e., since evolution is random, if the max point is has sharp peaks then it might be more evolutionary stable to go at a slightly lower peak but a flatter peak.

8. Half concentration Impacts

Here we have the K_x and K_y and then I found the maximal value of p, η for each. The color scale is the fitness from blue at lowest to red at highest.

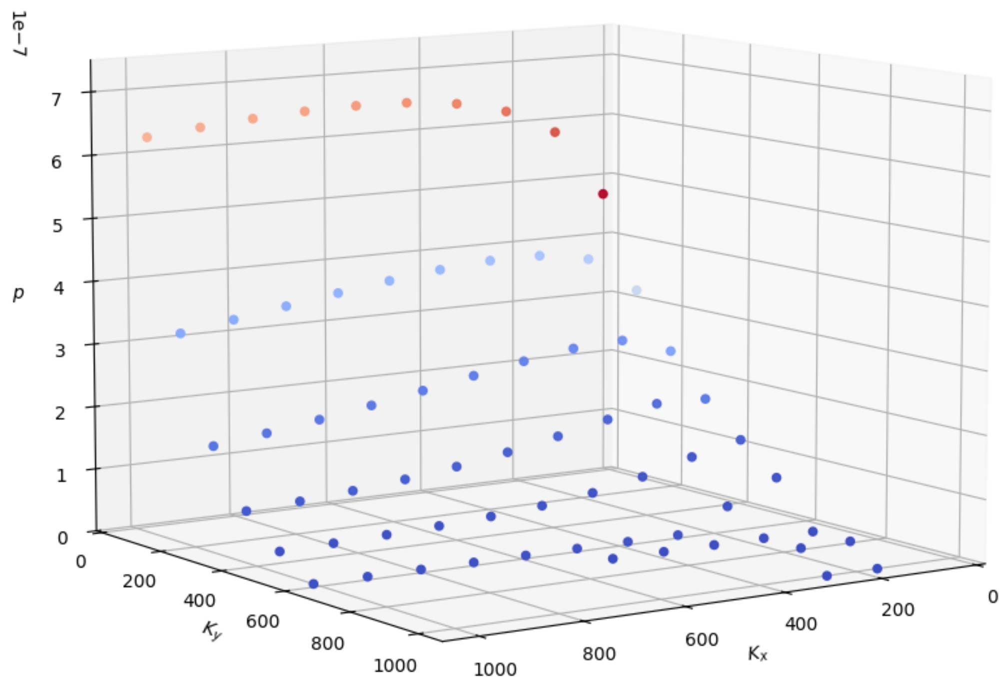

Figure 5: Here we see that K_y largely determines the value of p, as $K_x \to 0$ the p value decreases a bit, however it is mostly determined by K_y

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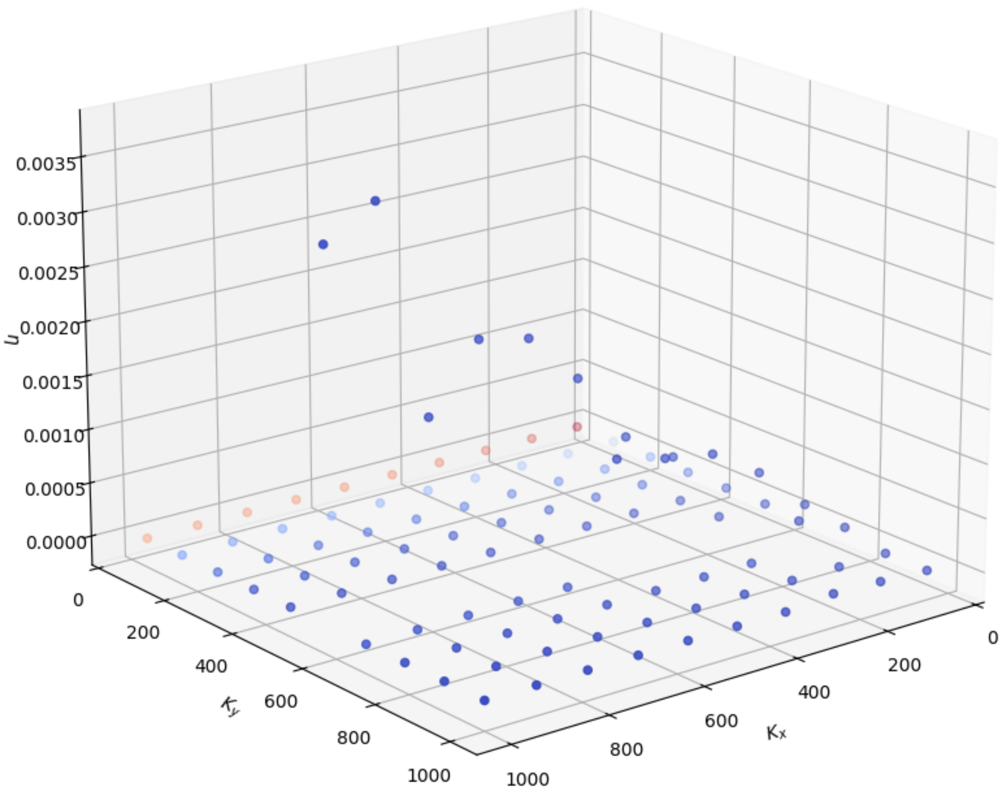

Figure 6: Here we see that K_x, K_y have an interesting effect on η

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$K_y = 600$</td><td>$K_x = 400$
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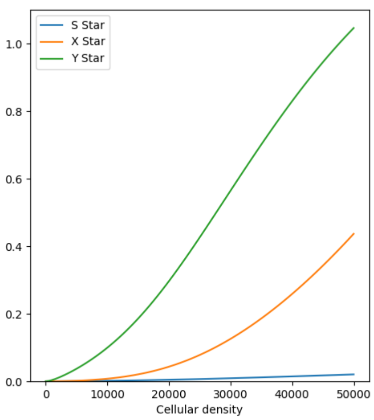
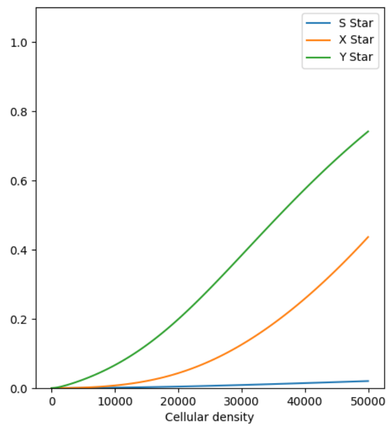
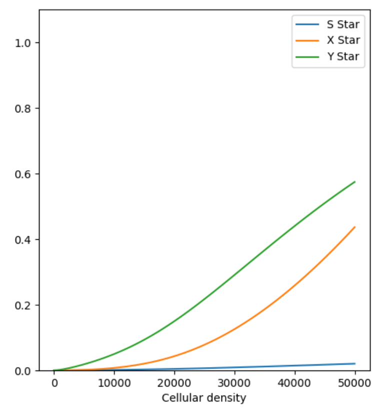
<tr><td>
</td><td>
</td><td>
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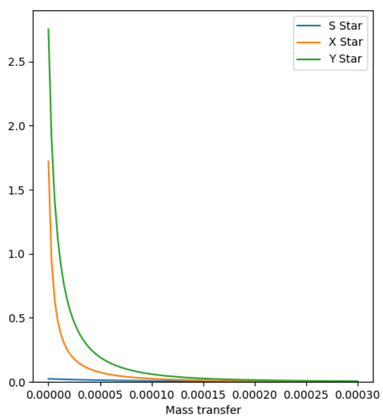
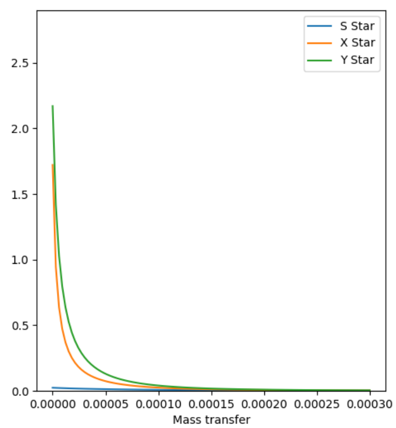
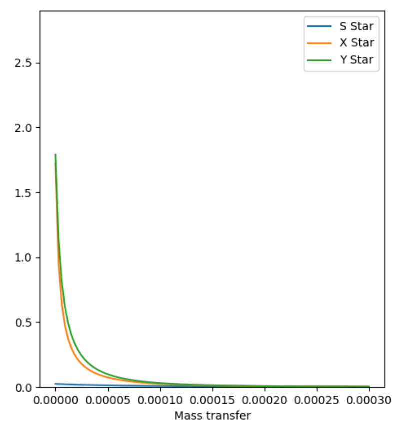
<tr><td>
</td><td>
</td><td>
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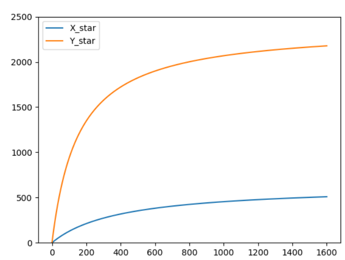
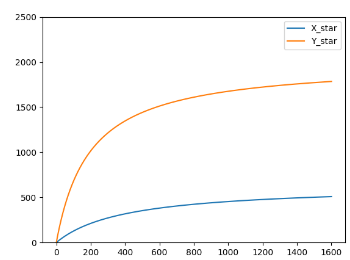
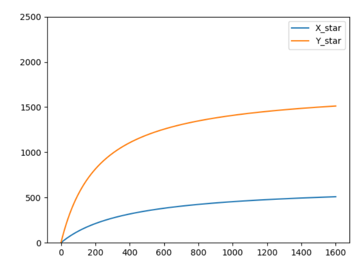
<tr><td>
</td><td>
</td><td>
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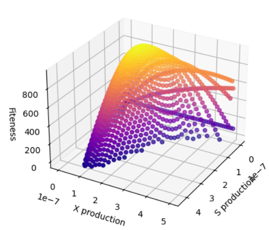
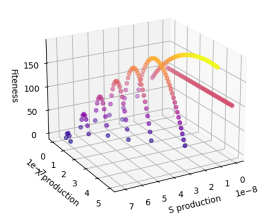
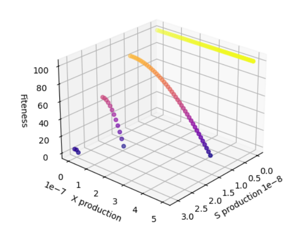
<tr><td>
</td><td>
</td><td>
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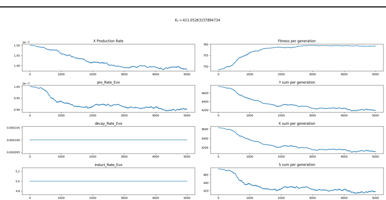
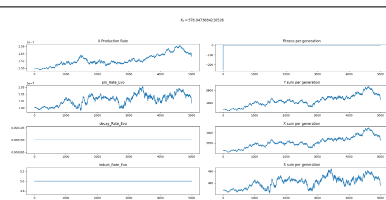
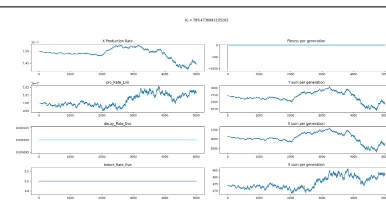
<tr><td></td><td></td><td></td></tr>
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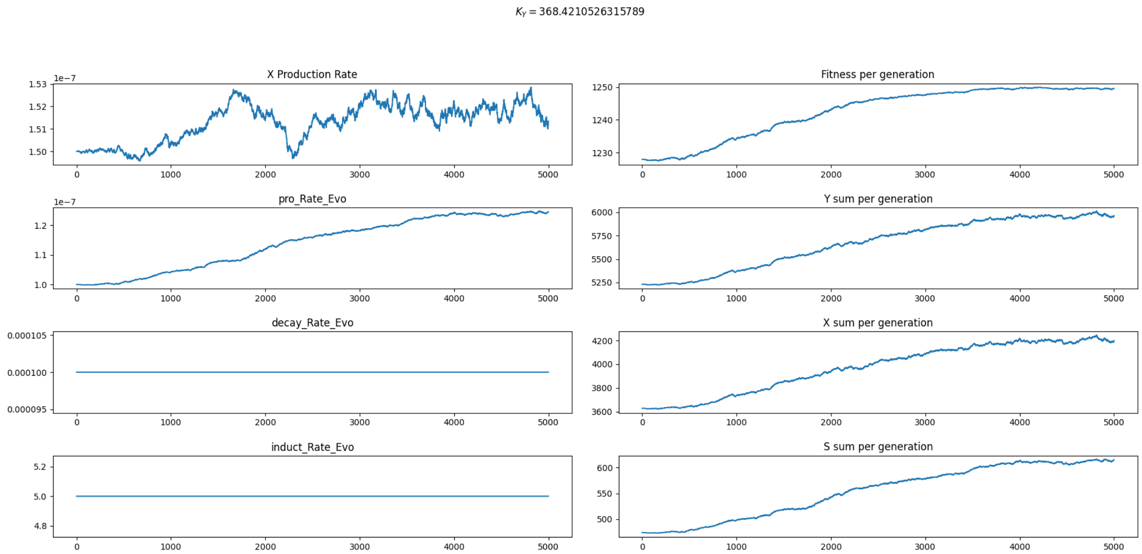
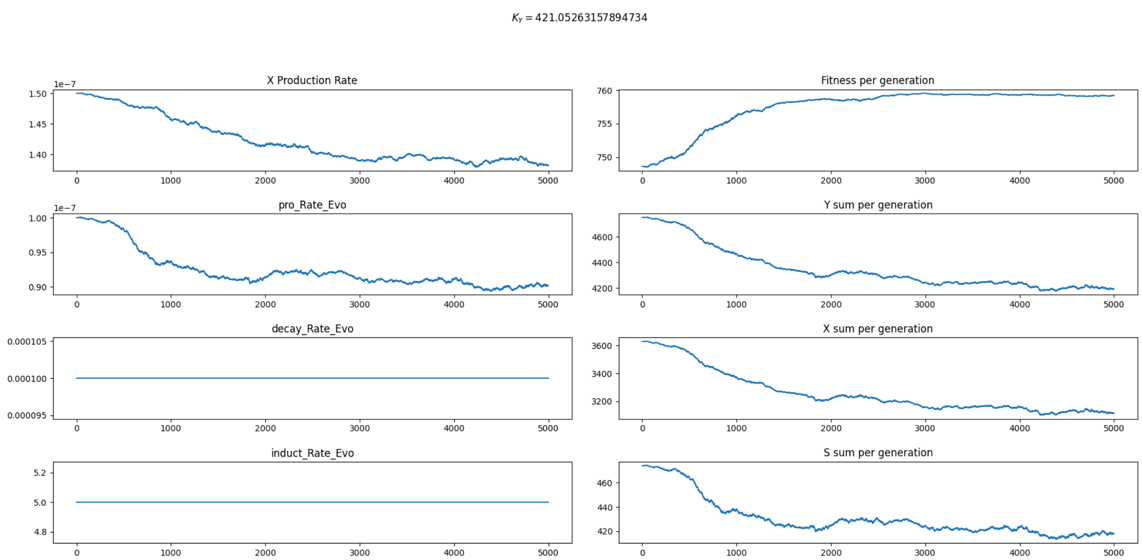
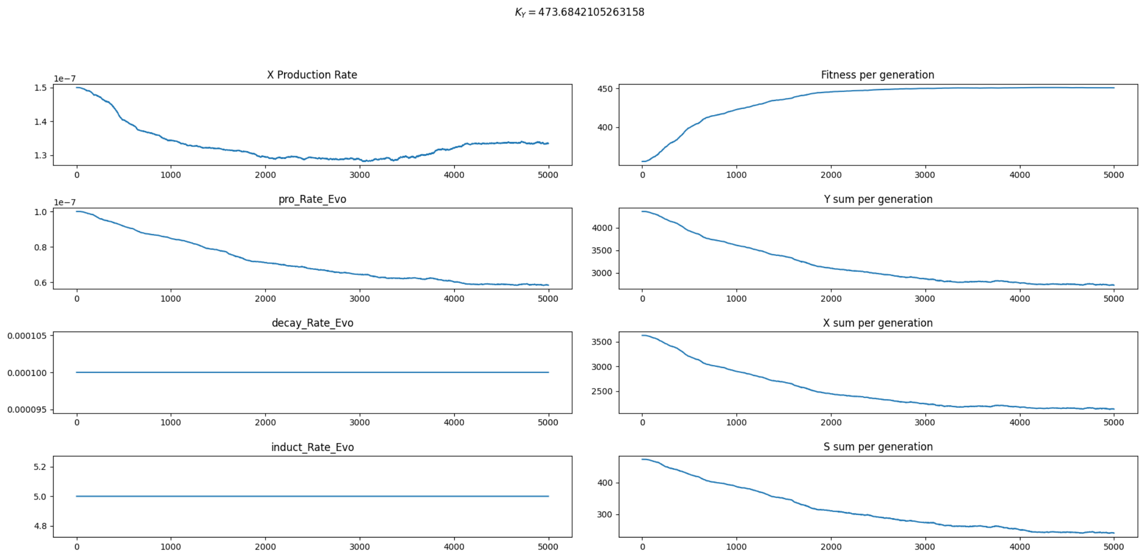
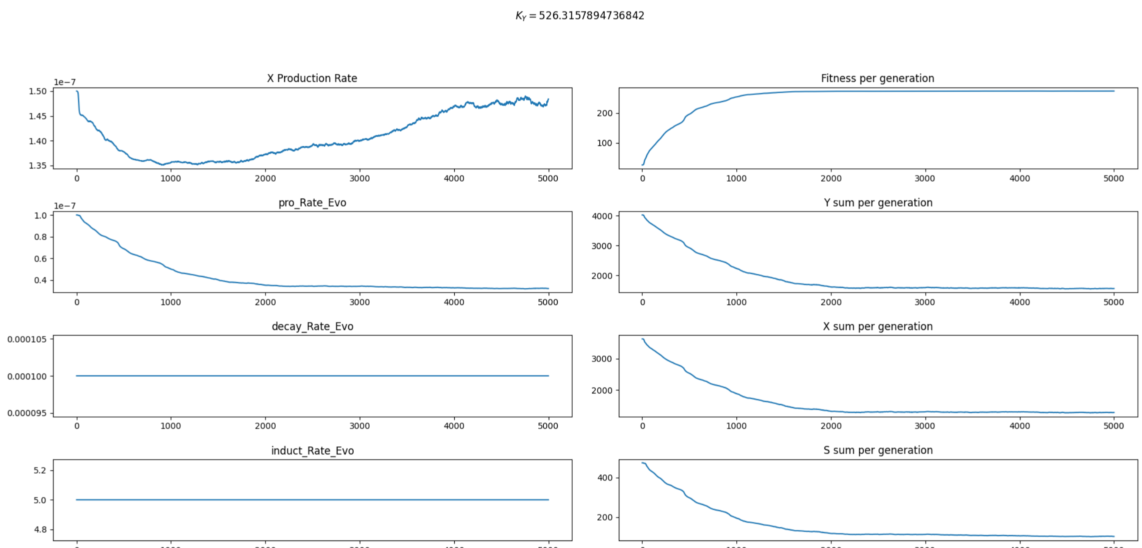
This explains the weird behavior of K_x and K_y on η and not p. We can see that the max p stays relativity the same for each K_y, however the reason that η changes so much is because of the tail end. At K_y = 600, the tail increases forever as we can see which is why η reaches such a high value. However we see at K_y = 400, the tail goes down rather than increase for η and so we get a very stable relative max. After about K_y = 600 we see that it kinda devolves with a lot of fitness being below the baseline of 100.



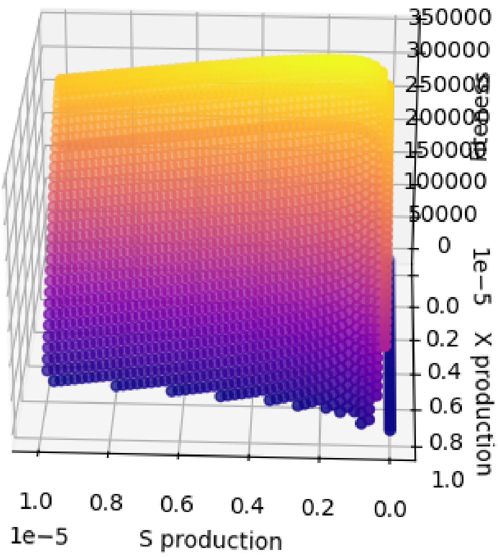




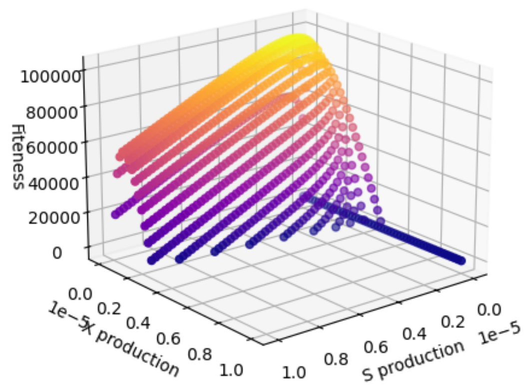


We see the geometry of the fitness function impact the results here. With $K_Y <\sim 450$, we see that the genotypes evolve to the global max, because one does exist, however if K_Y get too low, then we see this interesting instability in η. However after $K_Y >\sim 450$ we see that it evolves to the local max and then carries on toward the tail we see in the p vs η vs fitness graph.

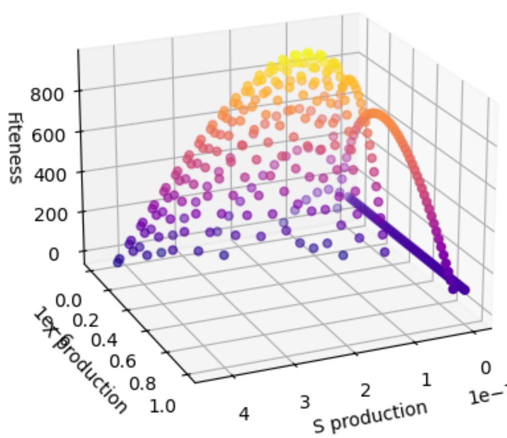
9. Effect of c



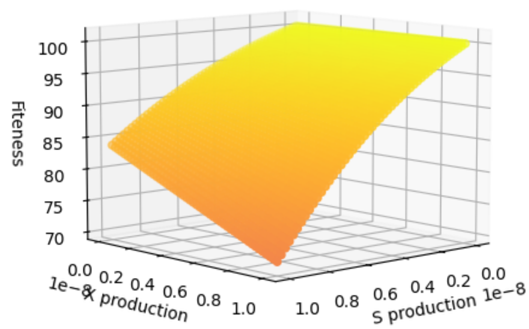
$c = 1e - 10$



$c = 1e - 9$



$c = 1e - 8$



$c = 1e - 7$

We can see that the c value flattens out the curve, it lowers the optimal p, η and also makes the tale a lot sharper, after a certain value it makes it so that the optimal is no communication at all, which we see at $c = 1e - 7$.

10. What has been done, What is next?

So far, most of the parameters have been tested, and the effects are noted here. We see that K_x, K_y affect the location of the local max and the ratio between them affects the tail of the fitness over S and X production. The Y consumption rate, c, affects the slope of the curve and the tails and local maxima of the graphs like the half signal concentration. It seems like c has a much larger impact more of the global qualities of the fitness graph and the half signal concentrations affect more of the local qualities of the graph. This is a bit simplified of an explanation but seems to be the case in the range of values that we would be testing over.

For the range of testing values, Mass transfer affects the curve much more that the cellular densities, at high mass transfers the impact is negligible, at lower mass transfers, higher cellular densities are more beneficial, which makes sense in the real world as well.

As for next steps, maybe testing some of the decay rates and X, Y interactions terms should be further investigated. probably in terms of ratios of the three as the mass transferred is added to them in every equation anyway. The auto induction term maybe should be looked at as well and maybe included into the evolving parameters as well. This was not done as we wanted to look at a more simplified model when validating it. Some actual testing, differing starting values and stability should be done as well. It could be interesting to choose a smaller range of mass transfer ranges and see how that impacts the evolved fitness function as well.